Reproducing and Extending a Paper on Particle Diameter Prediction

Your Name

Your Institute

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Problem Statement

- Particle diameter is critical in biomedical micro/nano systems.
- Full experimental exploration is costly and time-consuming.
- Goal: predict particle diameter from process parameters using ML.

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Paper Overview

- Target: diameter prediction for chitosan and TiO2-related particles.
- Baseline approach: ANN models (MLP and RBF).
- Inputs: process parameters; Output: particle diameter.

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What I Reproduced

- Rebuilt data workflow and model pipeline from the paper.
- Implemented baseline training and evaluation steps.
- Compared reproduced metrics with paper-reported performance.

Key Message

The original method was implemented as closely as possible.

Reproduction Results

- Baseline results follow the trend reported in the paper.
- Any small gap can come from split randomness and implementation details.
- Reproduction validates the paper's methodology in this environment.

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My Extensions

- Added a new model beyond the original paper.
- Added a new dataset to test broader applicability.
- Added an optimization phase for better tuning.

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Comparison: Baseline vs Extended Pipeline

- **Paper baseline** $\rightarrow$ reproduced.
- + **New model** $\rightarrow$ improved candidate behavior.
- + **New dataset** $\rightarrow$ better robustness check.
- + **Optimization** $\rightarrow$ stronger final performance.

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Conclusion

- I reproduced the paper's core workflow and results pattern.
- I extended the work with a new model, new dataset, and optimization.
- The pipeline is both reproducible and improvable.

Takeaway

This project validates the original study and shows practical next-step improvements.

Thank You

Questions?